MODUL 14. STATIC MODIFIER, INHERITANCE, POLIMORPHISM

Capaian Pembelajaran Praktikum:

- Menggunakan static modifier
- Menerapkan konsep inheritance dalam program
- Menerapkan konsep polimorfisme dalam program

Tools:

- Java Development Kit (JDK)
- Eclipse

Terminologi:

Isikan terminology yang sesuai untuk definisi dibawah ini:

[.....................................] Is a keyword that makes a variable, method, or inner class available without first creating an instance of a variable.

[.....................................] The concept in object-oriented programming that allows classes to gain methods and data by extending another classes fields and methods.

[.....................................] A programming philosophy that promotes protecting data and hiding implementation in order to preserve the integrity of data and methods.

[.....................................] A keyword that allows subclasses to access methods, data, and constructors from their parent class.

[.....................................] Visible to the package where it is declared and to subclasses in other packages.

[.....................................] Classes that are more specific subsets of other classes and that inherit methods and fields from more general classes.

[.....................................] Implementing methods in a subclass that have the same prototype (the same parameters, method name, and return type) as another method in the superclass.

TRY IT / SOLVE IT:

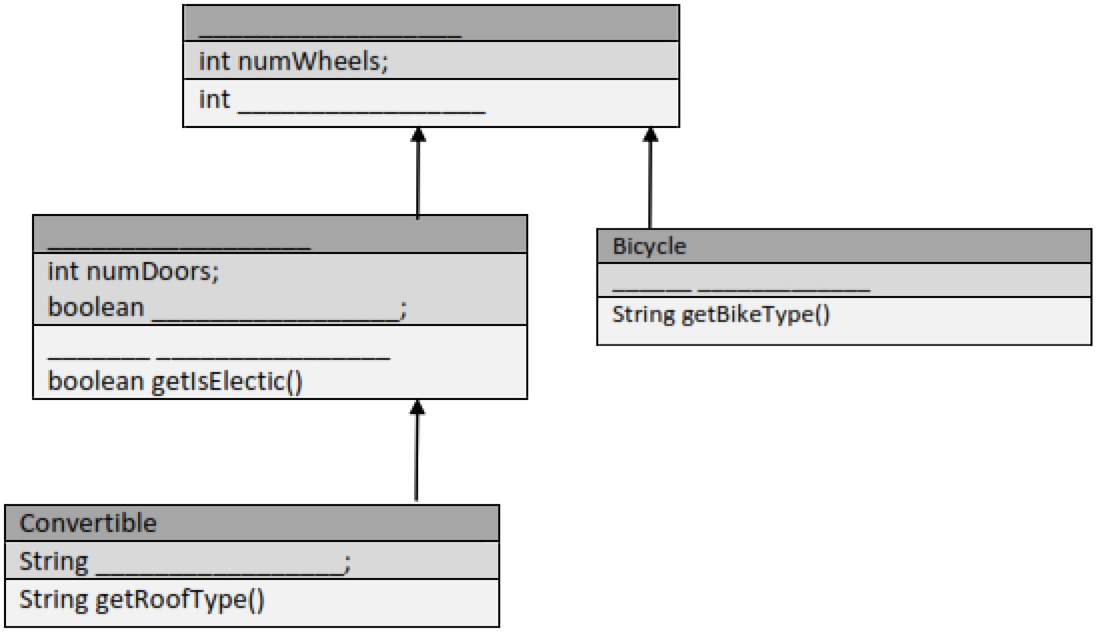
1. Buat sebuah class Turtle dan class DriverTurtle sbb: Amati kesalahan/error yang terjadi pada class DriverTurtle, jelaskan penyebab error dan coba anda perbaiki.

```java
public class Turtle {
    public static String food = "Turtle Feed";
    private int age;
    private int tankNum;
    public static int numTanks = 3;
    public Turtle(int age){
        this.age = age;
        tankNum = (int)((Math.random()*numTanks)+1);
    }
    //public void swim(){}
    public int getAge(){
        return age;
    }
    public int getTankOfResidence(){
        return tankNum;
    }
    public static String fishTank() {
        return "I have " + numTanks + " fish tanks.";
    }

}
```

```java
public class DriverTurtle {
    public static void main(String[] args) {
        Turtle T1=new Turtle(1);
        Turtle T2=new Turtle(2);
        System.out.println("Jumlah Tangki Total adalah "+Turtle.numTanks);
        Turtle.numTanks=5;
        System.out.println("Turtle T1 berusia "+T1.age+" bulan");
        System.out.println("Turtle T1 berada di tangki nomor "+T1.tankNum);
        System.out.println(T1.fishTank());
        System.out.println("");
        System.out.println("Turtle T2 berusia "+T2.age+" bulan");
        System.out.println("Turtle T2 berada di tangki nomor "+T2.tankNum);
        System.out.println(T2.fishTank());
    }
}
```

2. Dengan menggunakan kode program dan UML di bawah ini, isi isian dengan keyword yang tepat. Kemudian tulis program dengan Eclipse.




```
public class Vehicle {
      __________ int numWheels;
      __________ Vehicle(int numWheels)
      {
            this.numWheels = numWheels;
      }

      public int getWheels() {
            return wheels;
      }
}

public class Car __________ Vehicle {
      __________ int numDoors;
      __________ boolean isElectric;

      public Car (int numWheels, int numDoors, boolean isElectric) {
            __________(numWheels);
            this.numDoors = numDoors;
            this.isElectric = isElectric;
      }

      public int getNumDoors() {
            return __________;
```

```
    }

    public boolean getIsElectric() {
        return isElectric;
    }
}

public class Bicycle _________ Vehicle {
    //Mountain bike, road bike, recumbent bike.. etc
    _________ String bikeType;

    public Bicycle(int numWheels, String bikeType) {
        super(numWheels);
        this.bikeType = _________;
    }

    public _________ getBikeType() {
        return bikeType;
    }
}

public class Convertible _________ Car {
    //Soft top or rag top, or hard top
    _________ String roofType;

    public Convertible(int numWheels, int numDoors, _________ isElectric, String roofType) {
        super(numWheels, _________, _________);
        this.roofType = roofType;
    }

    _________ String getRoofType() {
        return roofType;
    }
}
```

3. Tulis program berikut kemudian jalankan! Beri penjelasan mengenai output yang didapatkan!

```
public class A {
    void callthis() {
        System.out.println("Inside Class A's Method!");
        }
}

public class B extends A{
    void callthis() {
        System.out.println("Inside Class B's Method!");
        }
}

public class C extends A {
    void callthis() {
        System.out.println("Inside Class C's Method!");
        }
}
```

```
public class DynamicDispatch {
    public static void main(String args[])  {
        A a = new A();
        B b = new B();
        C c = new C();
        A ref;
        ref = b;
        ref.callthis();
        ref = c;
        ref.callthis();
        ref = a;
        ref.callthis();
        }
}
```

LATIHAN:

4. Diberikan class Shape, Rectangle, Circle, Triangle, Square dan Object. Gambarkan diagram UML serta tuliskan kode programnya dengan definisi sbb:
    a. Square adalah sebuah Rectangle
    b. Rectangle, Triangle dan Circle merupakan tipe Shape
    c. Shape memiliki field color bertipe String dan method getColor() serta method hitungLuas().
    d. Rectangle memiliki field tambahan yakni panjang dan lebar bertipe int.
    e. Circle memiliki field tambahan yakni jejari bertipe int.
    f. Triangle memiliki field tambahan alas dan tinggi bertipe int.
    g. Rectangle, Circle dan Triangle memiliki method setter dan getter terhadap semua field.
    h. Rectangle, Circle dan Triangle melakukan override terhadap hitungLuas() dengan definisi luas menyesuaikan bentuknya.
    i. Square memiliki method setSisi() untuk mengatur nilai panjang dan lebar dengan isian yang sama.

5. Berdasarkan kode program poin 4, tuliskan kelas driver untuk membuat objek dan menerapkan konsep polimorfisme.

Setelah sesi praktikum SELESAI, laporan praktikum harus dikirim/diupload ke google classroom pada hari yang ditentukan.